

Cancer Classification using Neural Network

Simple project using sklearn + TensorFlow/Keras

Step 1: Import libraries

```
import numpy as np
import pandas as pd
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
```

Step 2: Load dataset

```
cancer = load_breast_cancer()
X = cancer.data
y = cancer.target
```

Step 3: Split data

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
                                                    random_state=42)
```

Step 4: Feature scaling

```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

Step 5: Build Neural Network

```
model = Sequential() model.add(Dense(16, input_dim=X.shape[1],  
activation='relu')) model.add(Dense(8, activation='relu'))  
model.add(Dense(1, activation='sigmoid'))
```

Step 6: Compile model

```
model.compile(loss='binary_crossentropy', optimizer='adam',  
metrics=['accuracy'])
```

Step 7: Train model

```
model.fit(X_train, y_train, epochs=50, batch_size=10,  
validation_split=0.1)
```

Step 8: Evaluate model

```
loss, accuracy = model.evaluate(X_test, y_test) print("Accuracy:",  
accuracy)
```

Step 9: Predict

```
predictions = (model.predict(X_test) > 0.5) print(predictions[:10])
```

Save model

```
model.save("cancer_model.h5")
```

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README.md (Add this to GitHub)

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Description

This project uses a neural network to classify whether a tumor is malignant or benign.

Dataset

Breast Cancer Dataset from sklearn

Technologies Used

- Python
- TensorFlow/Keras
- Scikit-learn

How to Run

1. Install dependencies: `pip install numpy pandas scikit-learn tensorflow`
2. Run the file: `python cancer_classification.py`

Output

- Model accuracy printed
- Predictions displayed

Author

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requirements.txt

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"""numpypandas scikit-learn tensorflow"""